ANNASETU · STAT REPORT AND CASE STUDY · ENGINE REPLAY

Three months of 2026, routed by the deployed engine

Every day from 1 March to 31 May 2026, simulated loads went through AnnaSetu's own advisor code on that day's real mandi data, at two scales.

Regional replay

Each day: ten tomato loads from a Kolar FPO, five 5-tonne loads from Azadpur, Delhi (tomato, onion, banana, cabbage, cauliflower) and one 2-tonne unsold tomato lot at Azadpur.

`1,380 loads · 3,772 t`

`92 unsold lots · 184 t`

Case study: one farm near Narela, Delhi

A simulated 8-acre farm: tomato on 3 acres, bottle gourd on 2, okra on 2 from mid-April, cauliflower on 1 in March; 300 kg of tomato back unsold each week. Its nearest reporting mandi is Azadpur, 17 km away.

`112 loads · 127 t`

`13 unsold lots · 3.9 t`

Money saved

Rs 2.20 cr

likely Rs 1.73-2.96 crore

Farm: Rs 5.2 lakh (3.7-7.7)

Kept out of landfill

180 t

likely 104-407 t; 169 t rescued + 11 t net waste avoided

Farm: 4,420 kg (2,149-11,389)

Advice held out of sample

97.6%

1,248 of 1,279 diverted loads

Farm: 85.9%, 79 of 92

Sent away from the nearest mandi

94.6%

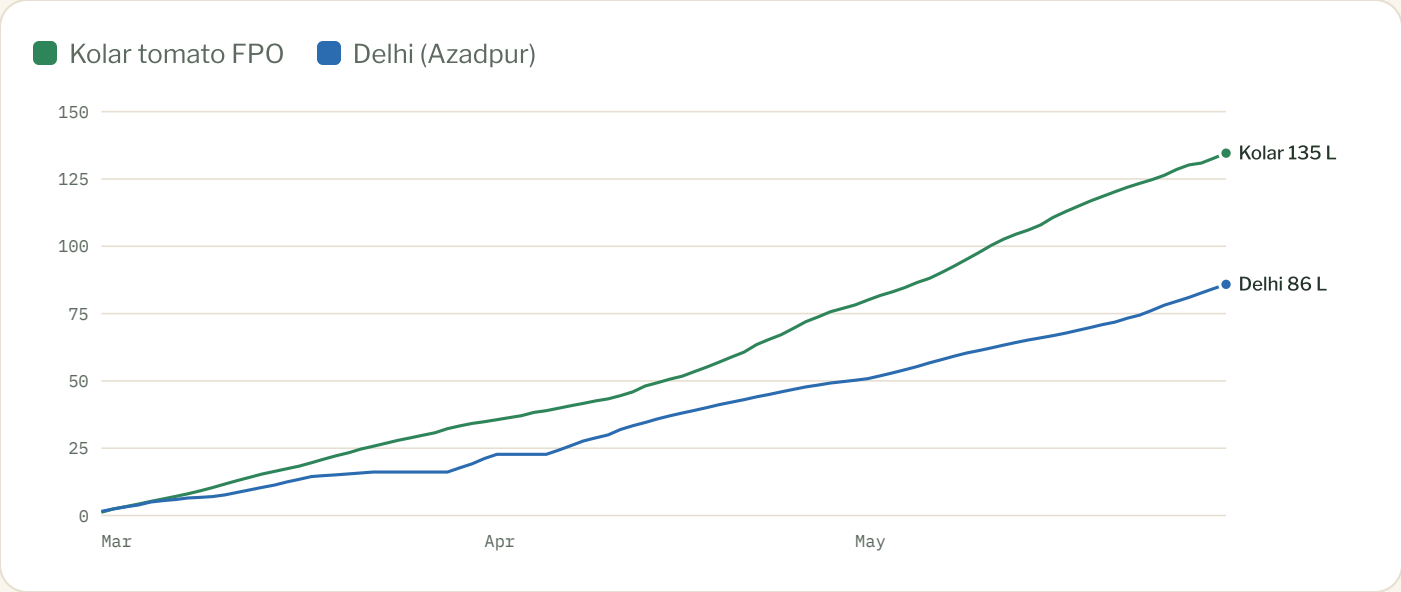
1,306 of 1,380 loads; +77,984 km in all

Farm: 85.7%, 96 of 112

RESULT

Money AnnaSetu added, day by day

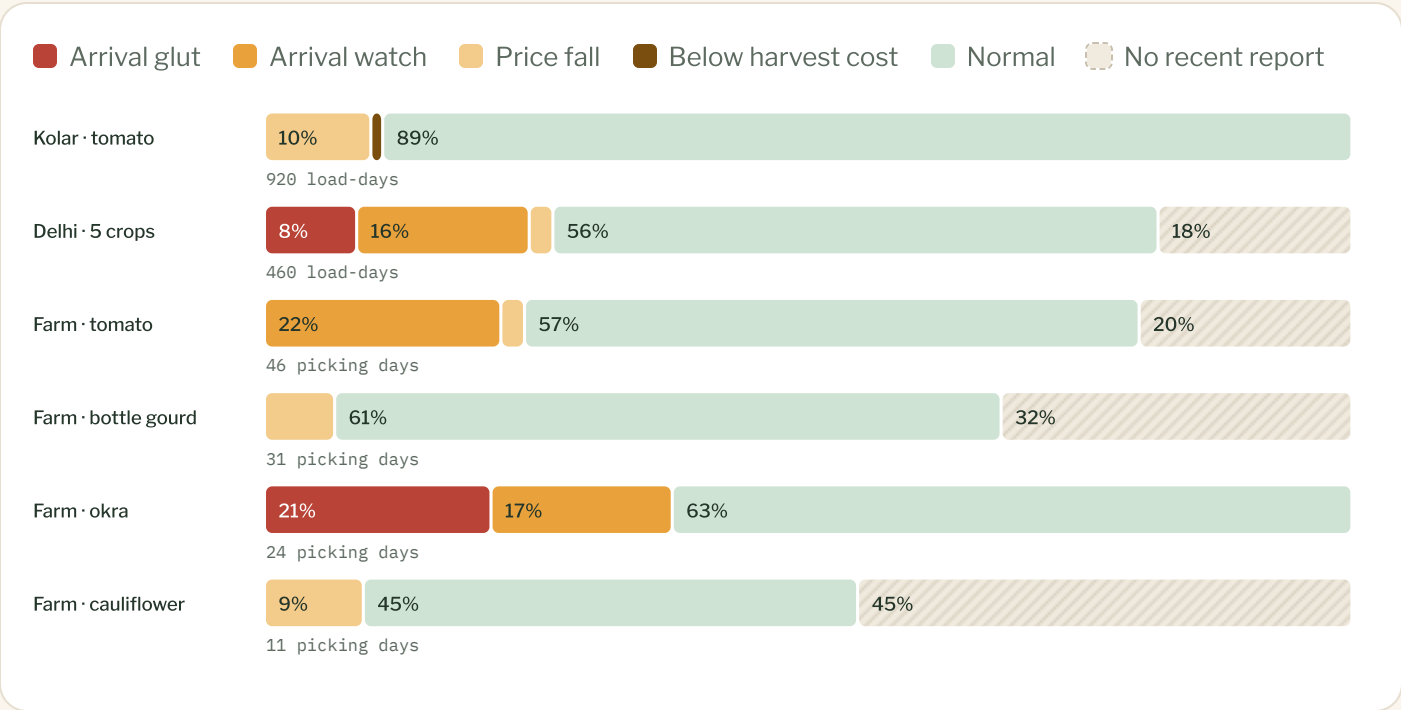
Regional replay, cumulative money saved (Rs lakh), mid estimate: expected earnings at the advised market minus the nearest mandi, after freight, fees and spoilage (Section 9).



PROBLEM CLASSIFICATION

What was wrong at the nearest mandi

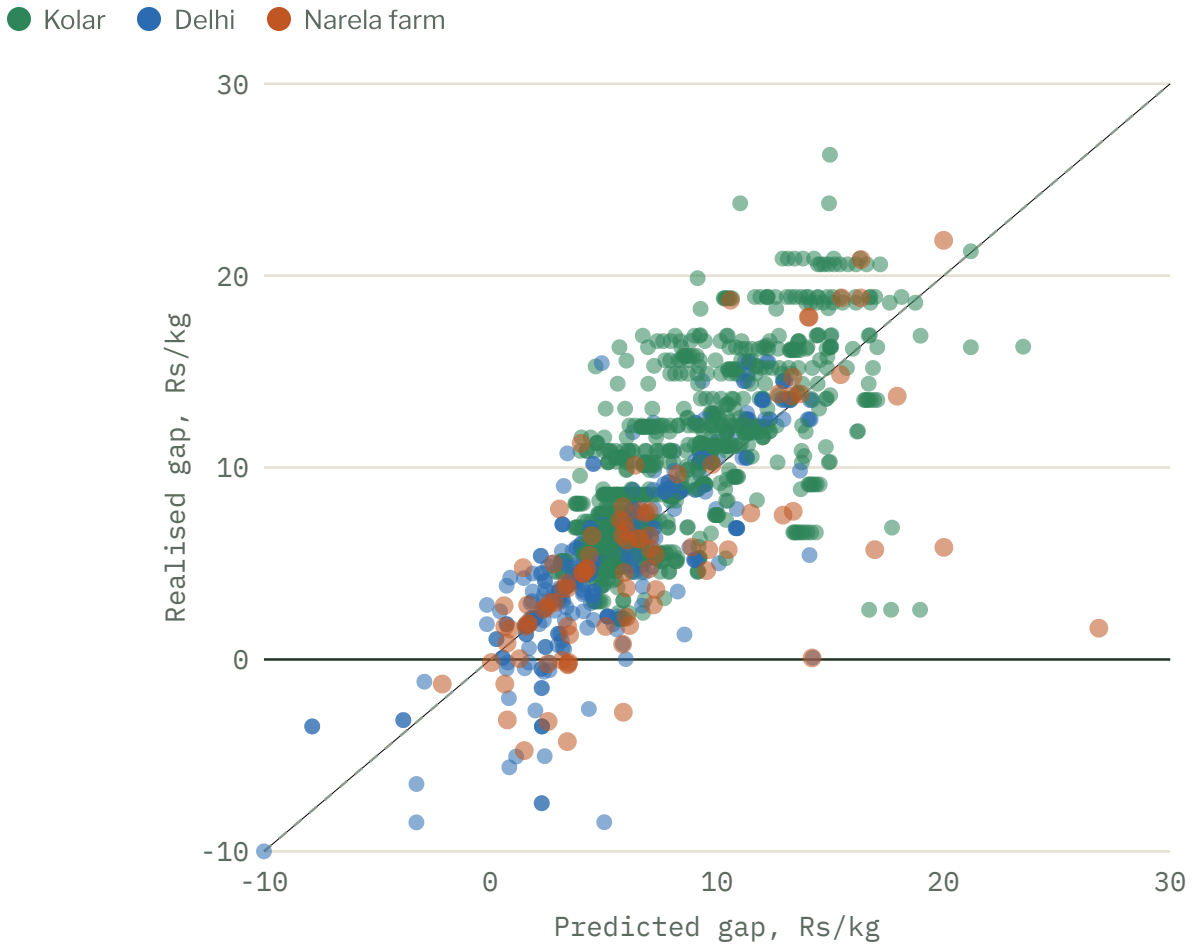
Each load-day is classed by the nearest mandi's state that day (Section 9 Step 1 rules; below cost per D12). Delhi's arrival gluts were all banana at Azadpur, and the farm's were okra after mid-April; Kolar had no arrival glut this spring, only price falls.



VERIFICATION

Did the advised market really pay more?

Out-of-sample check on every load sent away from the nearest mandi: the advised and the nearest market's next real price report (within 7 days after the decision), minus freight. Above the zero line, the advice held. The diagonal is a perfect prediction.



RESULT BY CROP

Gains come from glut days and price gaps, not every load

On normal days a longer trip adds spoilage, so waste avoided can be negative even when the price is better. The app never shows a load's negative figure (D35); this table does. Cauliflower is the weak spot at both scales.

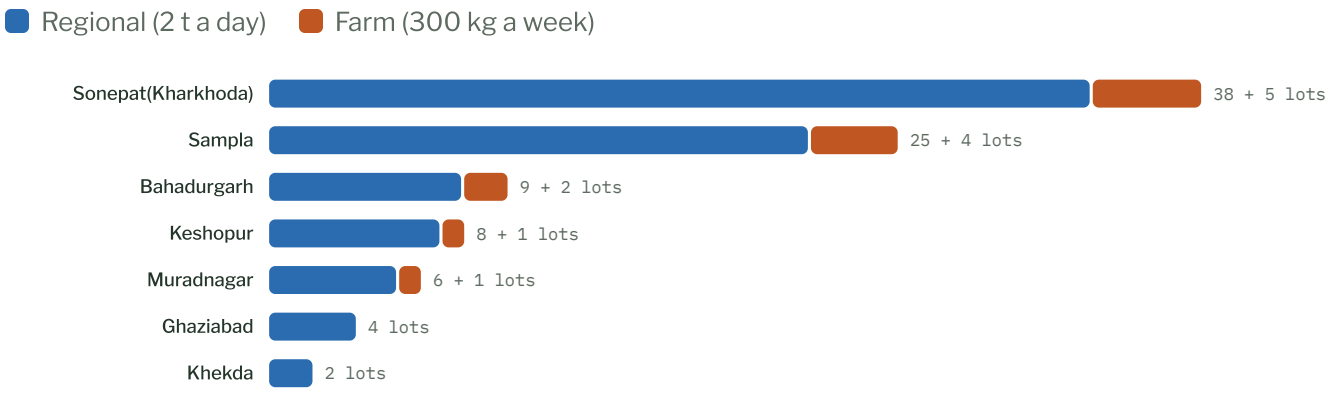
CROP	LOADS	SENT ELSEWHERE	ADVICE HELD	REALISED GAP	WASTE AVOIDED, KG	MONEY SAVED
Kolar FPO						
Tomato	920	920	903/903	Rs 9.19/kg	-6,466 (-27,436 to +82,914)	Rs 134.6
Delhi, Azadpur						

CROP	LOADS	SENT ELSEWHERE	ADVICE HELD	REALISED GAP	WASTE AVOIDED, KG	MONEY SAVED
Banana	92	84	82/82	Rs 10.51/kg	+26,437 (-12,498 to +78,837)	Rs 35.(
Cabbage	92	92	90/90	Rs 6.84/kg	+1,343 (-3,407 to +30,843)	Rs 25.:
Tomato	92	81	69/79	Rs 4.02/kg	-114 (-10,326 to +29,168)	Rs 13.:
Onion	92	92	90/90	Rs 3.25/kg	-7,564 (-9,202 to +14,324)	Rs 10.4
Cauliflower	92	37	14/35	Rs -0.49/kg	-2,366 (-2,366 to +2,384)	Rs 1.9
Narela farm						
Tomato	46	46	40/44	Rs 4.54/kg	+378 (-979 to +4,787)	Rs 2.5
Bottle gourd	31	21	18/21	Rs 5.84/kg	-57 (-172 to +608)	Rs 0.7
Okra	24	19	17/17	Rs 5.85/kg	+750 (-50 to +2,270)	Rs 1.8
Cauliflower	11	10	4/10	Rs -0.26/kg	-227 (-227 to +148)	Rs 0.

RESULT · RESCUE

Unsold stock at Azadpur found a buyer every time

The engine proposed the spoiled share from that day's temperature and sent the edible part to the best-paying mandi within 100 km (D36). Lots sent to each mandi, regional and farm together.



Rescued: 169 t of 184 t regional (91.8%), 3,576 kg of 3,900 kg from the farm (91.7%); the rest is the estimated spoiled share. No biogas or compost unit near Azadpur took it (D26).

CLASSIFICATION OF EVERY FIGURE

What is observed, modelled or assumed

FIGURE	STATUS	SOURCE
Mandi prices and arrivals, 2021 to May 2026	Observed	AGMARKNET market-level files, India Data Portal (D24)
Realised prices used for verification	Observed	Same files, reports after each decision day
Hourly temperature at Kolar, Azadpur and Narela	Observed	NASA POWER (MERRA-2), D16
Road distance and drive time	Observed	Amazon Location Service routes, cached
Glut ratio R, risk level, problem class	Model	Section 9 Step 1 on observed arrivals and prices
Net value per kg, money saved	Model	Section 9 Steps 2-4, 6
Spoilage on the trip, edible/spoiled split	Model	Step 3: shelf life, Q10, temperature, hours
Waste avoided, kept out of landfill	Model	Step 6, ranges
Freight Rs 11/tonne-km, commission, hamali, fees	Assumption	Desk research, placeholders until FPO interviews (Section 10)
Dump share u(R), price-below-cost dump	Assumption	Placeholder table (Section 9, D12)
The loads, the unsold lots, the farm and its picking quantities	Simulated	Fixed daily set; farm quantities from typical yields for the area, not one grower's records

VERIFICATION OF THE BUILD

Checks behind these numbers

- ✓ **Same code as production.** Both runs call the advisor's own /plan and /recommend functions; only the date and the data source change.
- ✓ **0 failed requests** in 1,562 (regional: 1,380 loads in 184 plans and 92 unsold lots; farm: 112 loads on 77 picking days and 13 unsold lots).

- ✔ **Deterministic.** Re-running 10-11 April gave identical output for all 30 loads; the engine has no randomness.
- ✔ **120 backend tests pass**, including the Section 17.3 cases (normal week, glut day, ten-load split, rescue, recover order).
- ✔ **Out of sample:** the advised market's next real price beat the nearest mandi's after freight on 1,248 of 1,279 diverted regional loads (Kolar 903/903, Delhi 345/376) and 79 of 92 farm loads.
- ! **Cauliflower:** the advice held on only 14 of 35 Delhi loads and 4 of 10 farm loads (median realised gap Rs -0.49 and -0.26/kg). Its prices at the advised markets moved more than their history suggested.
- ! **Kolar sent every load away** because Bengaluru paid Rs 7-9/kg more after freight all spring; the price edge was real (903 of 903 held), but more trucks to one city would test the anti-herding cap, which Live mode cannot check without arrivals (D34).
- ! **Predicted gaps ran high on the farm's okra and bottle gourd** (median Rs 9.63 and 7.19/kg predicted, 5.85 and 5.84 realised); the direction held, the size was smaller.

All money and waste figures are estimates (D32). Loads, lots and the farm are simulated; prices, arrivals, weather and routes are real. Reproduce: `python analysis/simulate_2026.py` and `python analysis/simulate_farm_2026.py`, then `python analysis/sim_summary.py` (with `--prefix farm_2026` for the farm).